

Activity 10.5: Building Geological Intuition with Numerical Experiments

Geophysics — Numerical Heat Flow

Purpose

To use numerical experiments to explore how thermal structure responds to geological complexity.

Thought Experiments

Consider modifying a numerical heat flow model to include:

- depth-dependent thermal conductivity,
- layered heat production,
- time-varying basal heat flux,
- transient magma intrusions.

Tasks

1. For each modification, predict qualitatively how the temperature field would respond.
2. Identify which changes could still be modeled analytically and which require numerical approaches.
3. Discuss how numerical models allow separation of physical effects that are difficult to isolate observationally.

Key Insight

Numerical modeling enables controlled experiments on Earth systems that cannot be performed in nature.